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Interpretable machine learning analysis of routine blood biomarkers and derived indicators for predicting coronary heart disease in patients with carotid stenosis

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Abstract

Background Coronary heart disease (CHD) and carotid stenosis share similar pathogenic mechanisms such as chronic inflammation, metabolic dysregulation, oxidative stress, and immune imbalance. These conditions are often comorbid. This study aimed to develop and validate an effective machine learning model that uses routine blood biomarkers to predict CHD risk in patients with carotid stenosis.

Methods Clinical data from 723 patients diagnosed with carotid artery stenosis between January 2019 and December 2024 were retrospectively collected, including demographic characteristics, hematological and biochemical laboratory parameters, and their derived composite indices. We used six feature selection methods and 11 machine learning (ML) algorithms to construct predictive models, and we systematically compared their performance. To interpret the established predictive model, we applied SHapley Additive exPlanations (SHAP) analysis to visualize and elucidate the risk prediction framework.

Results We identified the intersection of six feature selection methods, screened a total of nine potential predictors related to CHD, and used them to construct a prediction model. Cross-validation results demonstrated that the predictive model based on the random forest (RF) algorithm achieved the best performance among all evaluated algorithms (AUC = 0.800; sensitivity = 0.729; specificity = 0.792; F1 score = 0.733). The SHAP plots for RF indicate that estimating residual cholesterol (RC), fasting plasma glucose (FPG), and monocyte-to-lymphocyte ratio (MLR) are the three most important features for predicting CHD. Significant differences were observed in SHAP values for key biomarkers (including RC, MLR) across varying degrees of carotid artery stenosis.

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Conclusions The RF model achieved the best predictive performance in predicting CHD in patients with carotid stenosis. Elevated metabolic and immunoinflammatory markers significantly enhanced the predictive power of the model. Prospective multicenter validation is warranted to confirm its generalizability across diverse populations.

Keywords Coronary heart disease, Carotid stenosis, Machine learning, Prediction model

Introduction

Ischemic stroke and acute myocardial infarction remain the leading causes of disability and mortality worldwide, with the disease burden of carotid stenosis and coronary heart disease (CHD) being particularly substantial [1]. The core pathophysiological foundation shared by both conditions is atherosclerosis, a systemic vascular disorder characterized by chronic inflammation [2–4].

Lipid-driven inflammatory cascades within the arterial wall regulate complex interactions between multiple cells and molecules, thus laying a common biological framework for the onset and progression of cardiovascular and metabolic diseases [3, 5]. In the dynamic structure of the atherosclerotic plaque, deposition of lipid metabolites, foam cell formation, immune cell infiltration, and sustained release of pro-inflammatory mediators combine to build a highly active pathological micro-environment [5]. When plaques become destabilized and ultimately rupture, they can precipitate catastrophic events, including cerebrovascular accidents and acute myocardial infarction, imposing severe medical burdens and socioeconomic consequence [1, 6].

Although coronary computed tomography angiography (CTA) and invasive coronary angiography provide precise anatomical information for the assessment of CHD, their use in population-based screening is limited by their invasive nature, high cost, operator dependence, and limited generalizability, especially in asymptomatic populations [7, 8]. More critically, such imaging modalities typically reflect only the static anatomy of focal stenosis, making it difficult to capture the key pathological processes that drive atherosclerosis progression on a systemic scale [2, 8, 9].

In contrast, hematological biomarkers based on routine laboratory tests offer a non-invasive, cost-effective, and scalable alternative for the assessment of these systemic pathological mechanisms. The integration of hematological and biochemical parameters holds the promise of identifying individuals at elevated risk for CHD before significant anatomical changes in the disease occur, thus complementing or even partially surpassing imaging-based risk assessment strategies alone [7, 10].

A growing body of evidence emphasises the increasing clinical importance of metabolic and immunoinflammatory markers as proxies for vascular inflammation, immune dysregulation, oxidative stress, and endothelial dysfunction [10–12]. Hematologic inflammatory ratios including the neutrophil/lymphocyte ratio (NLR),

monocyte/lymphocyte ratio (MLR), platelet/lymphocyte ratio (PLR), and metabolic parameters, such as the triglyceride-glucose index (TyG) and residual cholesterol (RC), have gradually emerged as surrogate biomarkers for atherosclerotic burden and overall cardiovascular risk [13–17]. These easily accessible indicators have been shown to correlate significantly with the occurrence and prognosis of cardiovascular disease and have demonstrated significant clinical value in diagnosis, monitoring and risk prediction [10, 16–18].

In this context, machine learning (ML) provides powerful computational tools to capture complex non-linear associations between multiple clinical variables and construct high-performing individualised risk prediction models [19, 20]. In particular, interpretable machine learning techniques (e.g., SHapley Additive exPlanations (SHAP)) are able to bridge the gap between model performance and clinical interpretability by providing fine-grained resolution of feature contributions at both the overall and individual levels, and making highly complex algorithmic predictions transparent [7, 21].

In addition, patients with carotid stenosis are inherently at high risk of developing CHD due to common atherosclerotic mechanisms, but there is a lack of dedicated risk stratification tools for this vulnerable population. Therefore, there is an urgent need to develop robust, interpretable, and clinically applicable ML models that can systematically integrate routine laboratory markers and be used to predict CHD risk in patients with carotid stenosis.

Accordingly, this study addresses the key gaps in existing research by constructing and validating an interpretable ML framework that integrates conventional blood biomarkers and their derived indices for predicting CHD risk in patients with carotid stenosis, and by explicitly demonstrating the interpretability of the model through SHAP analysis. Given that early identification of CHD is essential for the prevention of stroke and myocardial infarction and for the necessary treatment of the brain and heart, the strategy of this study has significant potential to facilitate accurate risk stratification and optimize the management of cerebral and cardiovascular diseases (Fig. 1).

Methods

Data collection and definitions

Patients with carotid stenosis who attended Tianjin Huanhu Hospital between January 2019 and December 2024

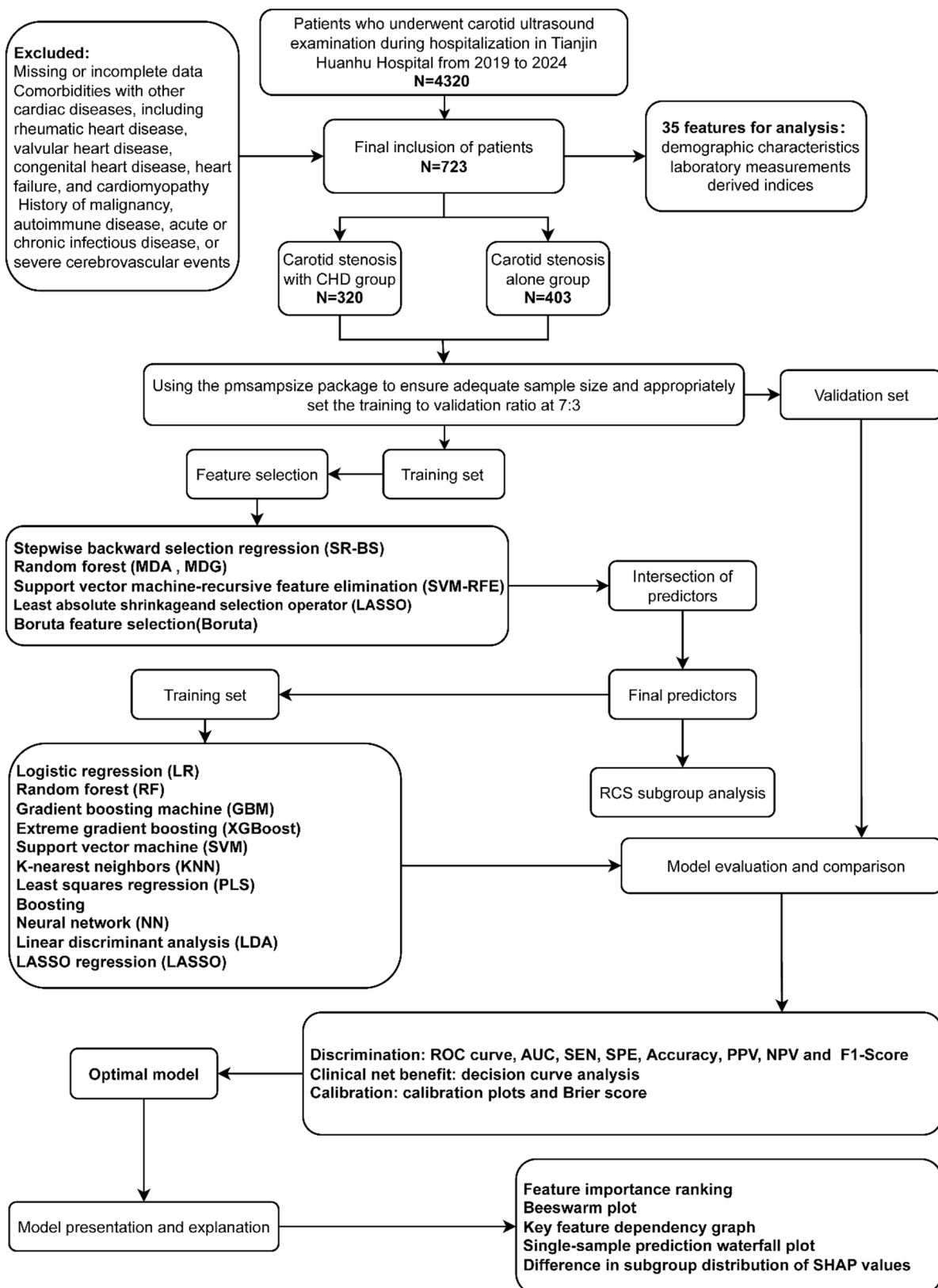


Fig. 1 Flowchart of the entire study. MDG, mean decrease Gini; MDA, mean decrease accuracy; ROC, receiver operating characteristic curve; SEN, sensitivity; SPE, specificity; AUC, area under the curve; SHAP, SHapley Additive exPlanations

were included in this study. From an initial cohort of 4302 patients who underwent carotid ultrasound during hospitalization, a total of 723 patients were included in the final analysis after screening according to inclusion and exclusion criteria. Participants must have ultrasound-confirmed carotid stenosis due to carotid atherosclerotic plaque. Based on the results of coronary CTA, all patients were categorized into a “carotid stenosis with CHD group” and a “carotid stenosis alone group”. Patients with missing or incomplete laboratory data were excluded from the study, as were those with a previous diagnosis of coronary artery disease or those with other cardiac diseases, including rheumatic heart disease, valvular heart disease, congenital heart disease, heart failure, and cardiomyopathy; patients with a history of malignant tumors, autoimmune diseases, acute or chronic infectious diseases, or severe cerebrovascular events were also excluded. This was a retrospective observational study, and all data were obtained from the hospital's electronic medical record system, with informed consent waived due to data anonymization. The study process strictly followed the relevant principles of the Declaration of Helsinki and was approved by the Ethics Committee of Tianjin Huanhu Hospital (Approval No. 2026-045).

We collected general clinical information about patients from electronic medical records. All laboratory tests were performed on admission as part of routine clinical assessment. We retained a total of 35 characteristics for analysis that included demographic characteristics, laboratory measures, and derived indices. These variables included sex, age, body mass index (BMI), history of diabetes, history of hypertension, history of smoking, and history of alcohol consumption, along with biochemical tests such as triglycerides (TG), total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), fasting plasma glucose (FPG), neutrophils, monocytes, lymphocytes, white blood cell count (WBC), red blood cell count (RBC), and platelet count (PLT).

We further calculated 15 metabolic and immunoinflammatory related composite indicators based on the basic biochemical parameters.

1. *Triglyceride-to-HDL-C ratio (THR)*: reflects the relative relationship between triglyceride levels and HDL-C, serving as a surrogate marker of dyslipidemia and insulin resistance. $THR = \text{triglycerides} \div \text{HDL-C}$.
2. *Residual cholesterol (RC)*: assesses cholesterol beyond LDL-C and HDL-C, representing a residual risk marker for atherosclerosis. $RC = TC - \text{LDL-C} - \text{HDL-C}$.
3. *Triglyceride-glucose index (TyG)*: a surrogate marker of insulin resistance widely used in cardiovascular

research. $TyG = \ln[(\text{triglycerides} \times \text{fasting plasma glucose}) \div 2]$.

4. *Neutrophil-to-lymphocyte ratio (NLR)*: an important indicator of systemic inflammation. $NLR = \text{neutrophil count} \div \text{lymphocyte count}$.
5. *Derived NLR (dNLR)*: an NLR derivative adjusted for total WBC count, more sensitively reflecting leukocyte subset changes. $dNLR = \text{neutrophil count} \div (\text{WBC} - \text{neutrophil count})$.
6. *Monocyte-to-lymphocyte ratio (MLR)*: evaluates chronic inflammation and immune regulation status. $MLR = \text{monocyte count} \div \text{lymphocyte count}$.
7. *Platelet-to-lymphocyte ratio (PLR)*: reflects platelet-driven inflammation and thrombotic activity. $PLR = \text{platelet count} \div \text{lymphocyte count}$.
8. *Platelet-to-neutrophil ratio (PNR)*: associated with acute inflammation and thrombotic burden. $PNR = \text{platelet count} \div \text{neutrophil count}$.
9. *Systemic immune-inflammation index (SII)*: comprehensively reflects systemic immune and inflammatory activation. $SII = (\text{platelet count} \times \text{neutrophil count}) \div \text{lymphocyte count}$.
10. *Systemic inflammation response index (SIRI)*: characterizes innate immunity-driven inflammatory responses. $SIRI = (\text{neutrophil count} \times \text{monocyte count}) \div \text{lymphocyte count}$.
11. *Aggregate index of systemic inflammation (AISI)*: multidimensionally integrates inflammatory cell parameters. $AISI = (\text{neutrophil count} \times \text{monocyte count} \times \text{platelet count}) \div \text{lymphocyte count}$.
12. *Neutrophil-to-HDL-C ratio (NHR)*: reflects the relative balance between inflammatory activity and HDL anti-inflammatory capacity. $NHR = \text{neutrophil count} \div \text{HDL-C}$.
13. *Monocyte-to-HDL-C ratio (MHR)*: characterizes monocyte inflammatory activity relative to HDL anti-inflammatory mechanisms. $MHR = \text{monocyte count} \div \text{HDL-C}$.
14. *Platelet-to-HDL-C ratio (PHR)*: comprehensively evaluates platelet activation and HDL protective function. $PHR = \text{platelet count} \div \text{HDL-C}$.
15. *Lymphocyte-to-HDL-C ratio (LHR)*: reflects the interaction between lymphocyte immune function and HDL. $LHR = \text{lymphocyte count} \div \text{HDL-C}$.

All laboratory test values were obtained from fasting blood samples collected within 7 days before carotid ultrasound and coronary CTA. Due to significant missing and unavailable data, variables such as glycated hemoglobin, coagulation indices, or physical activity could not be included, which may confound associations between variables. The degree of carotid stenosis and plaque characteristics were assessed by experienced ultrasound professionals using a MyLab Resona 7 ultrasound system.

Scans included the common carotid artery, internal carotid artery, and carotid bifurcation. The degree of stenosis was classified as normal, < 50%, 50–69%, ≥ 70% to near occlusion, near occlusion, and complete occlusion. No stenosis or complete occlusion was excluded; stenosis < 70% was defined as mild-moderate stenosis and ≥ 70% as severe stenosis. Stable plaques were defined as those with homogeneous, smooth, regular surfaces with high echo intensity; unstable plaques were defined as hypoechoic or heterogeneous plaques based on plaque echodensity, or as fibrous cap incomplete plaques or ulcerated plaques based on plaque morphology [22]. Coronary heart disease (CHD) was defined as luminal stenosis ≥ 50% in at least one major coronary artery (left anterior descending, left circumflex, and right coronary arteries) [16].

Feature selection

Before determining the final predictors, we cleaned the data and estimated the sample size with the *pmsampsize* package to ensure a sufficient sample size and to set the ratio of the training set to the validation set as 7:3. We then randomly divided the dataset into the training set and validation set according to the set ratio, and we used the training set to perform feature screening.

To enhance model robustness and interpretability, this study integrated six complementary feature selection strategies to evaluate variable importance from multiple dimensions and reduce redundancy. We employed the following feature selection methods.

1. *Stepwise Backward Selection Regression (SR-BS)*: Based on stepwise regression strategy, SR-BS eliminates variables with lower contribution according to statistical significance in each iteration.
2. *Least Absolute Shrinkage and Selection Operator (LASSO)*: By applying L1 regularization to regression coefficients, LASSO shrinks coefficients of less important variables to zero, constructing a more parsimonious feature set. Its advantage lies in simultaneously achieving variable selection and model complexity control, thereby reducing overfitting risk.
3. *Random Forest (RF)*: Utilizing ensemble evaluation with multiple decision trees, RF can handle nonlinear relationships and interactions between variables. We employed two variants of the random forest method: mean decrease accuracy (MDA) and mean decrease Gini (MDG).
4. *Support Vector Machine-Recursive Feature Elimination (SVM-RFE)*: Utilizing its iterative elimination strategy, SVM-RFE optimizes feature subsets according to model performance, ensuring relevance to the classifier.

5. *Boruta Feature Selection (Boruta)*: An all-relevant feature selection method that constructs shadow features as baselines and repeatedly compares whether the importance of each real feature significantly exceeds that of shadow features. Repeatedly confirmed important features are retained, thereby avoiding omission of potentially effective variables during selection and serving as an important complement to other methods.

Variables identified by at least four methods were ultimately selected, and the final predictor set was determined by integrating clinical practice and expert opinion.

Model development and evaluation

Eleven machine learning (ML) algorithms were employed for model development on the training set, with the validation set used for model evaluation. ML algorithms included logistic regression (LR), random forest (RF), gradient boosting machine (GBM), extreme gradient boosting (XGBoost), support vector machine (SVM), k-nearest neighbors (KNN), partial least squares regression (PLS), boosting, neural network (NN), linear discriminant analysis (LDA), and LASSO regression (LASSO). Grid search with five-fold cross-validation was used to obtain optimized parameters.

Using the DALEX package (v2.5.3) to evaluate model performance, we assessed and compared discrimination, calibration, and clinical net benefit performance of all models. Multiple discrimination metrics included sensitivity and specificity, and area under the curve (AUC). Positive predictive value (PPV), negative predictive value (NPV), and F1-score were calculated. DeLong's test was used to compare AUC values among models. Decision curve analysis (DCA) was used to evaluate clinical utility and net benefit. The RF model was selected as the optimal model for subsequent analyses.

To evaluate model interpretability, we calculated SHAP values from the RF model using TreeSHAP. SHAP values represent log-odds of individual contributions and were visualized using the R package *shapviz* (v0.10.3) to interpret each feature's contribution to CHD risk prediction at both local and global levels.

Statistical analysis

Continuous variables were assessed for normality using the Shapiro-Wilk test. Normally distributed data were expressed as mean ± standard deviation, and non-normally distributed data were expressed as median and interquartile range. Categorical variables were expressed as frequencies and percentages. Comparisons between the two groups were made using the independent samples t-test, Mann-Whitney U-test, chi-square test, or Fisher's exact test. The Mann-Whitney U test was used

to compare the differences in the distribution of SHAP values for each variable between patients with mild-moderate and severe stenosis. A restricted cubic spline (RCS) model was constructed based on multivariate logistic regression (4 nodes at the 5th, 35th, 65th, and 95th percentiles), and heterogeneity of curve shapes was tested by introducing a “characteristics $\times$ degree of stenosis” interaction term. A likelihood ratio test of $P < 0.05$ indicated that the nonlinear association pattern was significantly different between subgroups, suggesting that the effect of this variable was moderated by stenosis severity; a $P < 0.05$ was considered statistically significant. The predictive ability of different immunoinflammatory indexes for coronary heart disease was evaluated using receiver operating characteristic (ROC) curve analysis and AUC. All data analyses were performed using IBM SPSS Statistics 23 and R Studio 4.5.2.

Results

Clinical and biological characteristics

A total of 723 patients were enrolled in this study, with a mean age of 66 years, of whom 82.2% were male. A total of 320 patients (44.3%) had carotid stenosis combined with CHD, and detailed baseline characteristics are shown in Table 1. Compared with patients without CHD, patients with carotid stenosis with CHD were older, had higher rates of smoking and alcohol consumption, and were more often affected by hyperlipidemia and diabetes mellitus. In terms of laboratory and biological parameters, patients with CHD showed higher levels of BMI, TG, TC, LDL-C, FPG, neutrophil count, monocyte count, and several inflammatory and lipid-derived parameters (including THR, RC, TyG, NLR, dNLR, MLR, PLR, PNR, SII, SIRI, AISI, and MHR). In addition, lymphocyte count and LHR levels were lower; all these differences were statistically significant. In addition, patients in the CHD group were more likely to have severe carotid stenosis and unstable plaques.

Selection and validation of predictor variables

The study randomly divided the entire sample into training and validation sets in a 7:3 ratio. The pooled distributions of the training and validation sets were relatively uniform. Six complementary variable screening methods were used on the training set, including SR-BS, Boruta, LASSO, RF-MDA, RF-MDG, and SVM-RFE. The variables obtained by each method are shown in Table 2. The six methods screened 17, 15, 12, 10, 10, and 7 candidate predictors, respectively. The intersection of the variables selected by the six methods is depicted in Fig. 2. The features jointly identified by at least four methods were selected as the final candidate variables. By combining clinical expert opinion with insights from the previous literature, nine variables were finally identified for

inclusion in the model development: RC, NLR, MLR, PLR, BMI, FPG, TyG, smoking, and drinking. Of these, TyG and RC were consistently identified in all six screening methods.

The screened variables were further analyzed for correlation and collinearity. Figure S1 shows a strong correlation between smoking and alcohol consumption ($r = 0.74$); TyG was moderately correlated with FPG ($r = 0.63$), NLR with MLR ($r = 0.47$) and PLR ($r = 0.65$), and MLR with PLR ($r = 0.45$). The results of the multicollinearity detection (Table S1) showed that the maximum value of VIF for all variables was 2.246, and no significant collinearity problems were found. Considering the robustness of the machine learning model to moderate correlation and the clear clinical significance of the variables, all variables were ultimately retained for modeling. The final analysis also confirmed that retaining this variable set did not reduce model stability or predictive performance.

Restricted cubic spline regression

The restricted cubic spline regression (RCS) model was used to examine the nonlinear associations between six metabolic and immunoinflammatory biomarkers and carotid stenosis severity. Among all variables analyzed (TyG index, RC, NLR, MLR, PLR, and FPG), FPG showed a significant non-linear association with CHD risk, moderated by carotid stenosis severity (p for interaction = 0.036) (Fig. S4). In the hypoglycemic state, the risk of CHD with severe stenosis was higher, whereas in the hyperglycemic state, the risk of CHD with mild-to-moderate stenosis increased significantly with increasing FPG, without exceeding that of those with severe stenosis.

SR-BS, stepwise regression-backward selection; Boruta, boruta feature selection algorithm; LASSO, least absolute shrinkage and selection operator; RF-MDA, random forest-mean decrease accuracy; RF-MDG, random forest-mean decrease Gini; SVM-RFE, support vector machinerecursive feature elimination; TG, triglyceride; HDL-C, high density lipoprotein-cholesterol; LDL-C, low density lipoprotein-cholesterol; PLT, platelet count; RC, residual cholesterol; TyG, triglyceride-glucose index; PLR, platelet-to-lymphocyte ratio; NHR, neutrophil-to-HDL-C ratio; PHR, platelet-to-HDL-C ratio; dNLR, derived neutrophil-to-lymphocyte ratio; BMI, body mass index; FPG, fasting plasma glucose; NLR, neutrophil-to-lymphocyte ratio; MLR, monocyte-to-lymphocyte ratio; SII, systemic immune-inflammation index; AISI, aggregate index of systemic inflammation; SIRI, systemic inflammation response index; LHR, lymphocyte-to-HDL-C ratio; MHR, monocyte-to-HDL-C ratio.

Table 1 Baseline characteristics of the whole data set

Characteristic	Total (n = 723)	Non-CHD(n = 403)	CHD (n = 320)	P-value
Age, y	66.00 (62.00, 71.00)	65.00 (61.00, 70.00)	67.00 (63.00, 72.00)	0.025
Sex, n(%)				0.682
Male	594 (82.16)	329 (81.64)	265 (82.81)	
Female	129 (17.84)	74 (18.36)	55 (17.19)	
Hypertension	584 (80.77)	321 (79.65)	263 (82.19)	0.390
Diabetes, n(%)	302 (41.77)	139 (34.49)	163 (50.94)	< 0.001
BMI, kg/m ²	22.92 (20.51, 24.74)	22.60 (19.85, 24.87)	23.34 (21.11, 24.64)	0.021
TG, mmol/L	1.27 (0.96, 1.68)	1.23 (0.90, 1.62)	1.38 (1.03, 1.79)	< 0.001
TC, mmol/L	3.64 (3.08, 4.31)	3.53 (3.02, 4.18)	3.75 (3.23, 4.46)	< 0.001
HDL-C, mmol/L	0.99 (0.82, 1.13)	0.99 (0.82, 1.13)	0.98 (0.82, 1.13)	0.640
LDL-C, mmol/L	2.19 (1.81, 2.69)	2.15 (1.77, 2.62)	2.26 (1.86, 2.78)	0.021
FPG, mmol/L	5.84 (5.29, 6.97)	5.67 (5.22, 6.65)	6.18 (5.39, 7.49)	< 0.001
WBC, 10 ⁹ /L	6.42 (5.47, 7.41)	6.39 (5.49, 7.36)	6.47 (5.45, 7.44)	0.625
Neutrophil, 10 ⁹ /L	3.85 (3.18, 4.75)	3.78 (3.16, 4.61)	3.93 (3.28, 4.86)	0.033
Lymphocyte, 10 ⁹ /L	1.85 (1.48, 2.23)	1.93 (1.58, 2.35)	1.76 (1.42, 2.15)	< 0.001
Monocyte, 10 ⁹ /L	0.38 (0.25, 0.50)	0.35 (0.22, 0.49)	0.40 (0.28, 0.50)	0.002
PLT, 10 ⁹ /L	209.00 (175.50, 240.00)	211.00 (177.50, 240.00)	206.00 (172.75, 239.25)	0.746
THR	1.32 (0.92, 1.90)	1.28 (0.84, 1.82)	1.40 (1.02, 2.00)	0.002
RC, mmol/L	0.41 (0.26, 0.58)	0.35 (0.23, 0.51)	0.49 (0.33, 0.67)	< 0.001
TyG	8.73 (8.40, 9.07)	8.65 (8.33, 8.96)	8.84 (8.46, 9.22)	< 0.001
NLR	2.07 (1.62, 2.64)	1.97 (1.53, 2.49)	2.25 (1.80, 2.83)	< 0.001
dNLR	1.52 (1.22, 1.91)	1.47 (1.18, 1.83)	1.62 (1.30, 1.97)	< 0.001
MLR	0.21 (0.13, 0.28)	0.19 (0.11, 0.26)	0.23 (0.15, 0.31)	< 0.001
PLR	112.18 (90.61, 141.99)	110.18 (88.40, 134.92)	118.16 (95.80, 151.14)	< 0.001
PNR	54.27 (43.57, 67.23)	56.65 (44.30, 67.91)	52.26 (42.27, 65.19)	0.043
SII	429.64 (326.32, 577.13)	405.88 (308.77, 526.12)	469.56 (346.38, 637.57)	< 0.001
SIRI	0.75 (0.46, 1.14)	0.67 (0.40, 0.99)	0.87 (0.56, 1.30)	< 0.001
AISI	158.88 (91.72, 237.94)	141.54 (80.47, 210.94)	183.43 (112.23, 268.46)	< 0.001
NHR	3.95 (3.03, 5.24)	3.83 (2.97, 5.10)	4.06 (3.14, 5.39)	0.064
MHR	0.38 (0.26, 0.52)	0.36 (0.22, 0.50)	0.40 (0.29, 0.54)	0.002
PHR	207.32 (169.98, 269.17)	210.84 (172.76, 261.48)	202.56 (164.69, 275.76)	0.647
LHR	1.91 (1.47, 2.47)	1.96 (1.55, 2.52)	1.87 (1.34, 2.34)	0.007
Hyperlipidemia, n(%)	251 (34.76)	116 (28.78)	135 (42.32)	< 0.001
Smoking, n(%)	288 (39.83)	127 (31.51)	161 (50.31)	< 0.001
Drinking, n(%)	271 (37.48)	119 (29.53)	152 (47.50)	< 0.001
Degree of carotid stenosis, n(%)				< 0.001
Mild-moderate	226 (31.26)	147 (36.48)	79 (24.69)	
Severe	497 (68.74)	256 (63.52)	241 (75.31)	
Carotid plaque stability, n(%)				0.026
Stable	385 (53.40)	230 (57.07)	155 (48.74)	
Unstable	336 (46.60)	173 (42.93)	163 (51.26)	

BMI Body mass index, *TG* Triglyceride, *TC* Total cholesterol, *HDL-C* High-density lipoprotein cholesterol, *LDL-C* Low-density lipoprotein cholesterol, *FPG* Fasting plasma glucose, *WBC* White blood cell count, *Neutrophil* neutrophil count, *Lymphocyte* Lymphocyte count, *Monocyte* Monocyte count, *PLT* Platelet count, *THR* Triglyceride to HDL-C ratio, *RC* Residual cholesterol, *TyG* Triglyceride-glucose index, *NLR* Neutrophil-to-lymphocyte ratio, *dNLR* derived neutrophil-to-lymphocyte ratio, *MLR* Monocyte-to-lymphocyte ratio, *PLR* Platelet-to-lymphocyte ratio, *PNR* Platelet-to-neutrophil ratio, *SII* Systemic immune-inflammation index, *SIRI* Systemic inflammation response index, *AISI* Aggregate index of systemic inflammation, *NHR* Neutrophil-to-HDL-C ratio, *MHR* Monocyte-to-HDL-C ratio, *PHR* Platelet-to-HDL-C ratio, *LHR* Lymphocyte-to-HDL-C ratio

Model development and comparison

We finally included nine variables. We split the training and validation sets in a 7:3 ratio, as in a previous study. The training and validation sets had similar distributions (Table S1). Only gender differed significantly ($P < 0.05$). The prevalence of CHD in the training set was 44.1%, and that in the validation set was 44.4%. We used 11

machine learning algorithms to build models. Table 3 lists validation set performance metrics. Model AUCs ranged from 0.725 to 0.801. DeLong test results showed that RF, GBM, SVM, and boosting had higher AUCs than others ($P < 0.05$ for each; Fig. S3). RF demonstrated the most balanced and superior predictive performance on the validation set. RF achieved the highest accuracy

Table 2 Predictor variables of carotid atherosclerosis in patients with CHD using different selection methods

Methods	Predictor variables
SR-BS	Diabetes, Smoking, Hyperlipidemia, Drinking, TG, HDL-C, WBC, LDL-C, Monocyte, PLT, RC, TyG, PLR, NHR, PHR, dNLR, Degree of carotid stenosis
Boruta	RC, BMI, TyG, FPG, NLR, MLR, SII, AISI, SIRI, PLR, Monocyte, Lymphocytes, LHR, MHR, dNLR
LASSO	Diabetes, Smoking, Hyperlipidemia, Drinking, Degree of carotid stenosis, FPG, RC, TyG, NLR, MLR, PLR, SIRI
RF-MDA	RC, Smoking, BMI, Drinking, TyG, FPG, NLR, MLR, SII, MHR
RF-MDG	RC, BMI, TyG, FPG, NLR, SII, AISI, MLR, SIRI, PLR
SVM-RFE	RC, TyG, TG, NLR, AISI, SIRI, MLR

SR-BS stepwise regression-backward selection, Boruta Boruta feature selection algorithm, LASSO least absolute shrinkage and selection operator, RF-MDA random forest-mean decrease accuracy, RF-MDG random forest-mean decrease Gini, SVM-RFE support vector machine-recursive feature elimination, TG triglyceride, HDL-C high-density lipoprotein-cholesterol, LDL-C low-density lipoprotein-cholesterol, PLT platelet count, RC residual cholesterol, TyG triglyceride-glucose index, PLR platelet-to-lymphocyte ratio, NHR neutrophil-to-HDL-C ratio, PHR platelet-to-HDL-C ratio, dNLR derived neutrophil-to-lymphocyte ratio, BMI body mass index, FPG fasting plasma glucose, NLR neutrophil-to-lymphocyte ratio, MLR monocyte-to-lymphocyte ratio, SII systemic immune-inflammation index, AISI aggregate index of systemic inflammation, SIRI systemic inflammation response index, LHR lymphocyte-to-HDL-C ratio, MHR monocyte-to-HDL-C ratio

(0.764), sensitivity (0.729), F1-score (0.733), and AUC of

0.800 (95% CI: 0.739–0.861), outperforming other algorithms across multiple evaluation metrics. Although GBM yielded a marginally higher AUC (0.801), RF exhibited more favorable overall discrimination and classification performance, particularly in sensitivity and F1-score, which are critical indicators for clinical screening scenarios where minimizing false negatives is prioritized. Furthermore, RF demonstrated acceptable calibration (linear slope = 1.205), indicating reasonable agreement between predicted probabilities and observed outcomes. These results suggest that the RF algorithm is the better machine learning model for predicting CHD in patients with carotid stenosis (Fig. 3).

Interpretation and interaction of variables in predictive outcomes

Figure 4A and B depict the relative contributions and directions of predictors to “carotid stenosis with CHD” predictions in the RF model. Ranked by mean |SHAP| values, RC, FPG, monocyte-to-lymphocyte ratio (MLR), BMI, neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), smoking, TyG, and drinking were most influential. SHAP value subgroup analysis (Fig. S4) revealed higher risk contributions in severe vs.

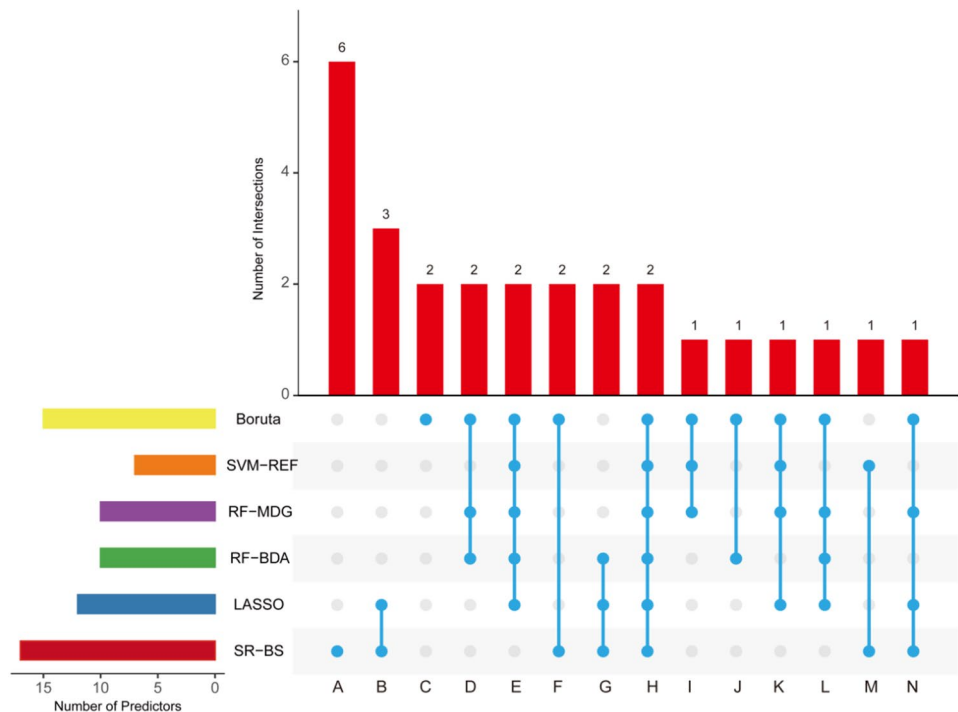


Fig. 2 Upset plot of interactions between the predictors. **A** high density lipoprotein-cholesterol, low density lipoprotein-cholesterol, white blood cell, platelet, neutrophil-to-HDL-C ratio, platelet-to-HDL-C ratio. **B** diabetes, hyperlipidemia. **C** lymphocytes, lymphocyte-to-HDL-C ratio. **D** systemic immune-inflammation index, body mass index. **E** neutrophil-to-lymphocyte ratio, monocyte-to-lymphocyte ratio. **F** monocyte, derived neutrophil-to-lymphocyte ratio. **G** smoking, drinking. **H** triglyceride-glucose index, residual cholesterol. **I** aggregate index of systemic inflammation. **J** monocyte-to-HDL-C ratio. **K** systemic inflammation response index. **L** fasting plasma glucose. **M** triglyceride. **N** platelet-to-lymphocyte ratio. Boruta, boruta feature selection algorithm, SVM-RFE, support vector machinerecursive feature elimination; RF-MDG, random forest-mean decrease Gini; RF-MDA, random forest-mean decrease accuracy; LASSO, least absolute shrinkage and selection operator; SR-BS, stepwise regression-backward selection

Table 3 Predictive performance metrics of different machine learning algorithms of the validation set

Models	AUC	Sensitivity	Specificity	Accuracy	PPV	NPV	F1
RF	0.800 (0.739–0.861)	0.729 (0.637–0.815)	0.792 (0.714–0.862)	0.764 (0.708–0.819)	0.737 (0.647–0.824)	0.785 (0.706–0.855)	0.733 (0.660–0.800)
GBM	0.801 (0.742–0.859)	0.708 (0.612–0.793)	0.742 (0.661–0.824)	0.727 (0.667–0.787)	0.687 (0.596–0.777)	0.761 (0.681–0.833)	0.697 (0.620–0.767)
SVM	0.785 (0.723–0.847)	0.625 (0.523–0.717)	0.800 (0.720–0.867)	0.722 (0.657–0.782)	0.714 (0.608–0.805)	0.727 (0.645–0.800)	0.667 (0.581–0.738)
LR	0.741 (0.675–0.807)	0.562 (0.458–0.663)	0.767 (0.685–0.841)	0.676 (0.606–0.741)	0.659 (0.549–0.757)	0.687 (0.604–0.765)	0.607 (0.514–0.688)
KNN	0.587 (0.513–0.661)	0.531 (0.429–0.630)	0.583 (0.496–0.675)	0.560 (0.491–0.630)	0.505 (0.409–0.602)	0.609 (0.513–0.696)	0.518 (0.426–0.599)
PLS	0.725 (0.656–0.793)	0.615 (0.511–0.708)	0.750 (0.670–0.824)	0.690 (0.625–0.750)	0.663 (0.560–0.761)	0.709 (0.624–0.786)	0.638 (0.550–0.715)
Boosting	0.781 (0.720–0.842)	0.656 (0.559–0.750)	0.767 (0.688–0.843)	0.718 (0.657–0.778)	0.692 (0.597–0.787)	0.736 (0.653–0.811)	0.674 (0.592–0.747)
NN	0.544 (0.467–0.621)	0.125 (0.062–0.191)	0.892 (0.835–0.946)	0.551 (0.481–0.616)	0.480 (0.280–0.682)	0.560 (0.487–0.629)	0.198 (0.104–0.288)
LDA	0.735 (0.668–0.802)	0.552 (0.449–0.650)	0.767 (0.685–0.841)	0.671 (0.602–0.736)	0.654 (0.544–0.753)	0.681 (0.599–0.759)	0.599 (0.506–0.684)
Lasso	0.741 (0.675–0.807)	0.552 (0.450–0.651)	0.758 (0.675–0.832)	0.667 (0.597–0.731)	0.646 (0.537–0.747)	0.679 (0.598–0.758)	0.596 (0.503–0.678)
XGBoost	0.766 (0.702–0.831)	0.698 (0.604–0.785)	0.758 (0.683–0.835)	0.731 (0.671–0.792)	0.698 (0.602–0.790)	0.758 (0.678–0.832)	0.698 (0.618–0.771)

RF Random forest, GBM Gradient boosting machine, SVM Support vector machine, LR Logistic regression, KNN K-nearest neighbors, PLS Partial least squares, Boosting boosting, NN Neural network, LDA Linear discriminant analysis, Lasso Least absolute shrinkage and selection operator, XGBoost extreme gradient boosting, PPV Positive predictive value, NPV Negative predictive value

mild-to-moderate stenosis for TyG ($P=0.011$), RC ($P=0.012$), NLR ($P=0.022$), MLR ($P=0.002$), PLR ($P=0.001$), and BMI ($P=0.029$). FPG ($P=0.203$), smoking ($P=0.100$), and drinking ($P=0.746$) showed no significant differences.

The SHAP dependency plot further revealed the effect patterns of the key variables and their potential interactions. Figure 5A shows that the contribution of RC shows a clear linear upward trend: as the value of RC increases, the SHAP value turns from negative to positive and continues to rise to about 0.2, suggesting that increased RC has a significant contribution to the risk of CHD. Scatter points colored by smoking status show a highly overlapping trend, suggesting limited modulation by smoking of RC's effect; only in the area of high RC values (>1 standard deviation) are slightly higher SHAP values observed in smokers than in nonsmokers, suggesting that there may be a mildly synergistic risk. Figure 5B shows that the relationship between FPG and SHAP values shows a nonmonotonic pattern. Within low to moderate levels of FPG (4.5–7 mmol/L), SHAP values gradually turn from negative to positive with increasing FPG, suggesting that elevated FPG is associated with a higher predicted risk of CHD. Alcohol consumption has a mild moderating effect on this relationship, with moderate drinkers having a reduced risk in the moderate glycemic range. In Fig. 5C, MLR shows a significant positive relationship with SHAP values. This positive relationship between MLR and SHAP values is evident, with SHAP values

continuing to increase as MLR increases and stabilizing at higher levels, suggesting that the incremental effect of MLR on disease risk is gradually saturated. The moderating effect of PLR on MLR is such that samples with a high PLR tend to have higher SHAP values, suggesting that inflammation and thrombosis may work together to drive an increased risk when the proportion of mononuclear cells is elevated and accompanied by increased platelet responses. BMI has a U-shaped effect on predicted risk (Fig. 5D). Low BMI corresponds to relatively high SHAP values, which may be associated with malnutrition, chronic inflammation, or insufficient metabolic reserve; when BMI is in the normal range, SHAP values cluster around zero, suggesting a low contribution to risk; and SHAP values increase again when BMI >25 kg/m², suggesting that overweight status increases disease risk through metabolic disruption mechanisms. Alcohol consumption has a limited effect on the risk contribution of BMI, which may only be slightly amplified in low BMI individuals. Figure 5E shows that NLR shows an overall trend of higher SHAP values with higher NLR, but the relationship is not strictly linear. Scatter point colors show that SHAP values are higher when TyG is also elevated, suggesting a mutually reinforcing effect between NLR (an indicator of inflammation) and TyG (a reflection of insulin resistance), with a more significant risk contribution when both are elevated at the same time. PLR also shows a non-linear effect (Fig. 5F). At low PLR levels, SHAP values are mostly negative or close to zero; as PLR

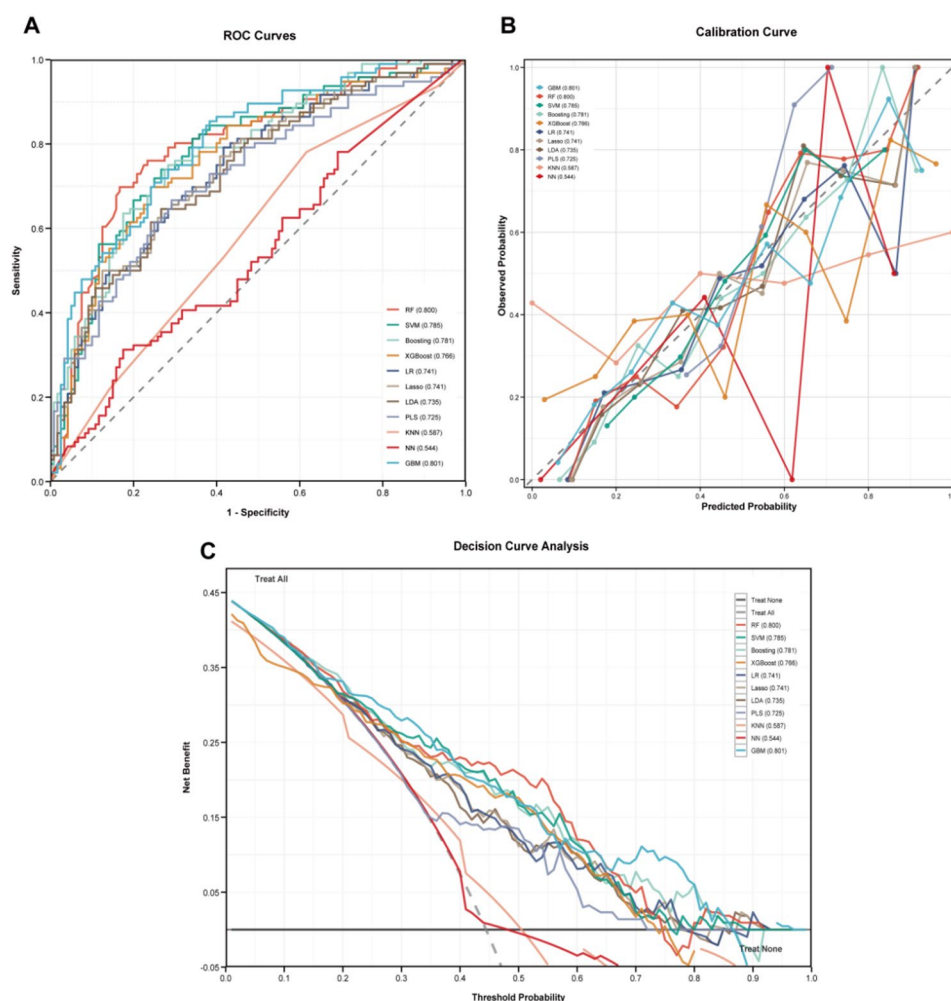


Fig. 3 ROC curves (A), calibration curves (B), and decision curve analysis (C) of machine learning models on the validation set. ROC, receiver operating characteristic curve; RF, random forest; GBM, gradient boosting machine; SVM, support vector machine; LR, logistic regression; KNN, k-nearest neighbors; PLS, partial least squares; NN, neural network; LDA, linear discriminant analysis; Lasso, least absolute shrinkage and selection operator; XGBoost, extreme gradient boosting; AUC, area under the receiver-operating curve.

increases, SHAP values increase rapidly; and at very high levels (> 300), the increase slows down but still maintains a high contribution. Figure 5G and I demonstrate the effect of alcohol consumption, TyG, and cigarette smoking on the model's predictions. The overall SHAP values of alcohol drinkers are higher than those of non-drinkers (Fig. 5G), suggesting that alcohol consumption is mainly risk-promoting and its effect does not depend on the level of FPG. TyG shows a non-linear increase in SHAP values (Fig. 5H), with individuals with high TyG showing a higher risk contribution; the SHAP values of smokers increase further when TyG increases, suggesting a possible synergistic effect. Figure 5I shows that smokers have significantly higher SHAP values than non-smokers; the risk contribution is more stable, and the effect does not change significantly with the level of other variables.

To demonstrate the model's prediction mechanism at the individual level, we analyzed typical high- and

low-risk subjects based on SHAP waterfall plots. Figure 4C shows that several metabolism and inflammation-related indices (e.g., RC, NLR, MLR, TyG, FPG) of the high-risk individual (predicted probability = 0.846) significantly and positively contributed to the prediction of the outcome, whereas the protective factors of non-smoking and non-alcohol consumption have limited contribution. In contrast, Fig. 4D shows that a low-risk individual (predicted probability = 0.114) presents low inflammation with more stable metabolism, and that the indicators such as MLR, NLR, TyG, and PLR collectively reduce the predicted values. Predictions matched actual outcomes for both. Overall, the model achieves an interpretable prediction of individual CHD risk by integrating multi-dimensional information on inflammation, lipid metabolism, and lifestyle factors.

In conclusion, through systematic SHAP analysis, this study not only identified the key predictors of carotid

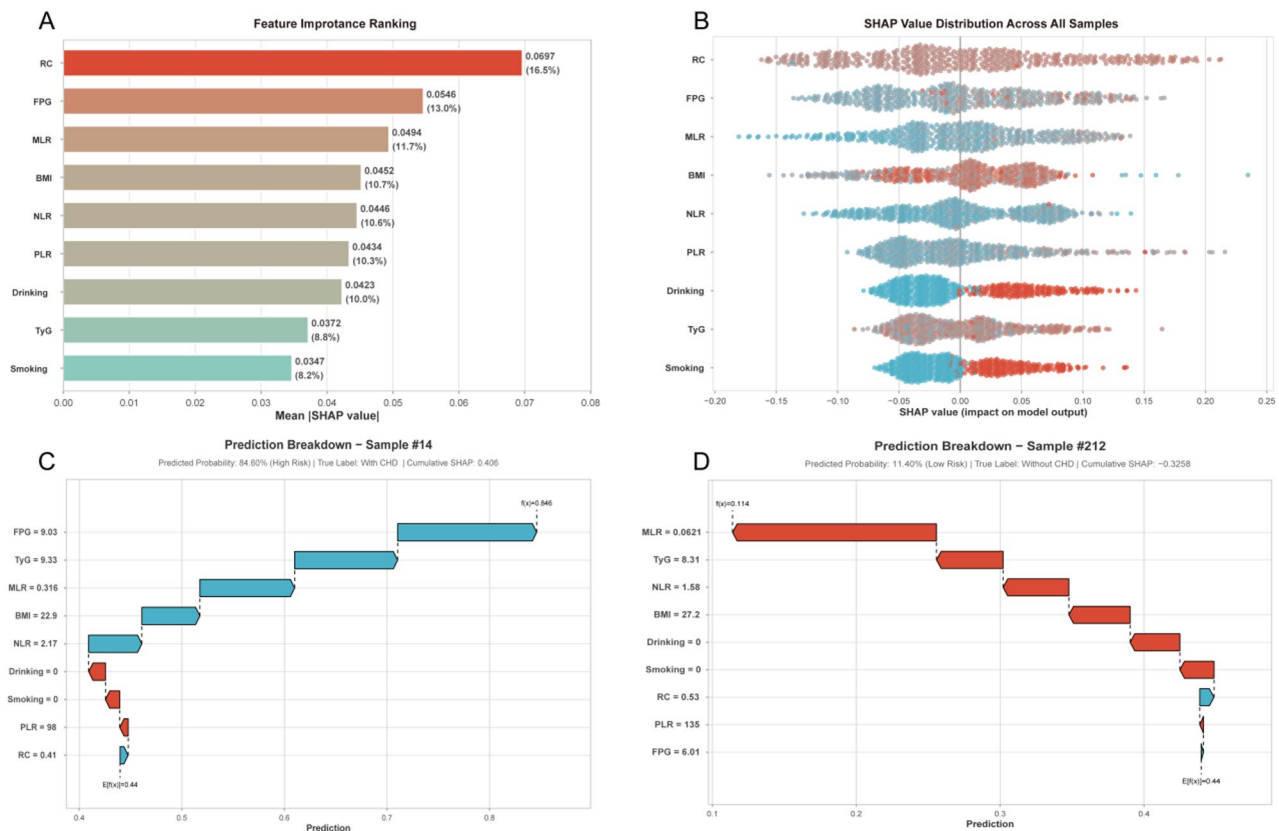


Fig. 4 SHAP Analysis of feature importance and individual prediction interpretation in coronary heart disease risk assessment. **A** The importance ranking of features based on the mean (|SHAP|). **B** SHAP value distribution across samples. Beeswarm plot showing individual SHAP values for each feature. Color gradient represents feature magnitude (red = high, blue = low). Horizontal dispersion indicates impact direction on CHD risk prediction. **C** High-risk case (Sample #14). Waterfall plot for confirmed CHD patient showing amplified risk (84.6% probability). **D** Low-risk case (Sample #212). Waterfall plot demonstrating feature contributions yielding 11.4% CHD probability. SHAP, shapley additive explanations; TyG, triglyceride-glucose index; RC, residual cholesterol; NLR, neutrophil-to-lymphocyte ratio; MLR, monocyte-to-lymphocyte ratio; PLR, platelet-to-lymphocyte ratio; BMI, body mass index; FPG, fasting plasma glucose

artery stenosis combined with coronary artery disease, but also revealed a complex network of interactions among metabolic indicators, immunoinflammatory indicators, and lifestyle factors. This interpretive analysis supports the biological validity of the model predictions and provides an effective basis for elucidating the disease mechanism and developing individualized precision prevention strategies.

Discussion

In this study, we combined six variable screening methods with real-world carotid stenosis data for the first time to develop a new approach to identifying risk factors for predicting CHD based on conventional clinical blood biomarkers and their derivatives. We evaluated 11 machine learning (ML) methods for prediction, with the RF model showing the highest performance. To gain deeper insight into the predictive mechanisms of the model and its potential clinical implications, this study primarily incorporated clinically applicable and readily identifiable predictors, while utilizing Shapley Additive

Explanations (SHAP) values to obtain more intuitive explanations. The SHAP dependency plot, by visualizing the complex relationship between predictor variables and model output, not only reveals the marginal effect of each feature on CHD risk prediction, but more importantly elucidates the pattern of interaction between features [19–21]. In this study, we performed a systematic dependency analysis on the most contributing features in the model. The results showed that metabolic and immunoinflammatory markers demonstrated differential non-linear prediction patterns; these findings provide a new perspective for understanding the pathophysiological mechanisms of comorbid coronary heart disease (CHD) in patients with carotid artery stenosis.

Residual cholesterol (RC) shows a significant positive correlation with CHD risk, and this effect is moderated by smoking status. Smokers exhibit higher SHAP values at the same RC level. Clinical studies have demonstrated that elevated RC levels are a strong predictor of cardiovascular events independent of LDL-C, and that this association persists in individuals with optimally

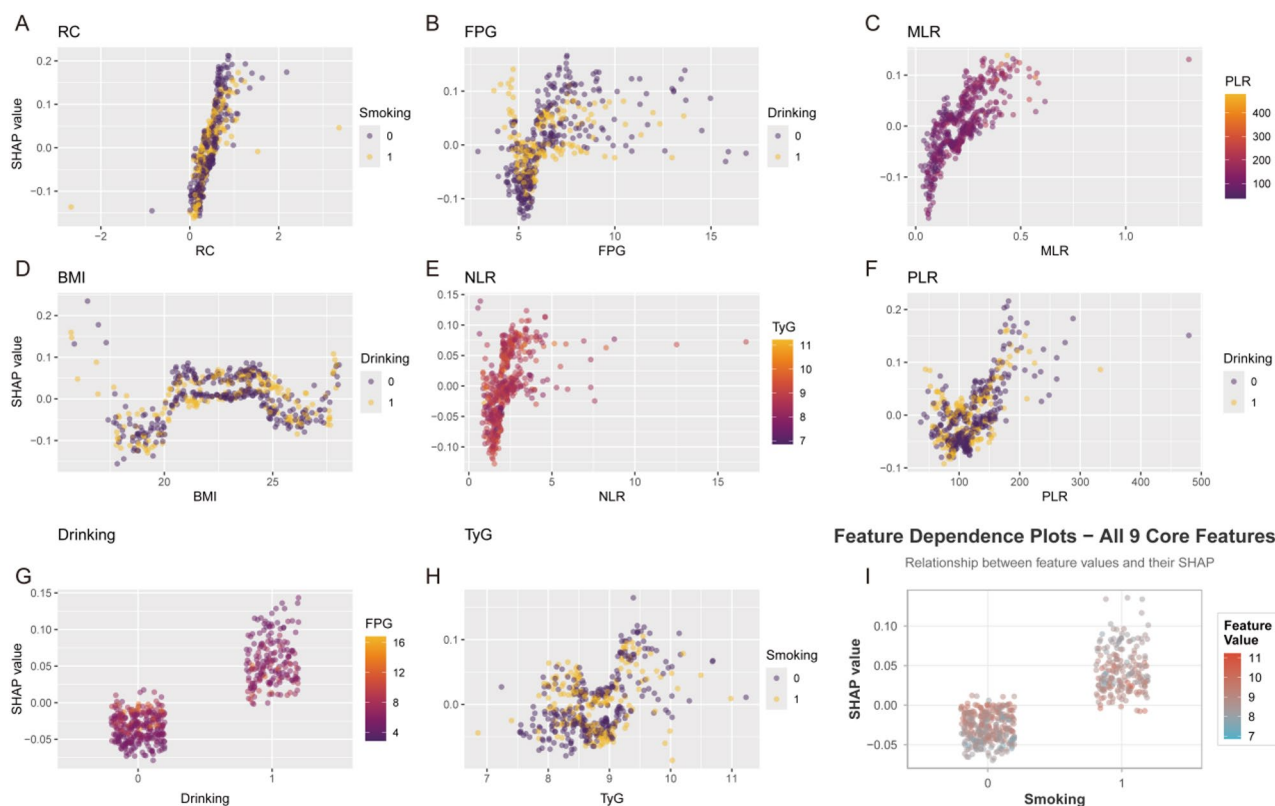


Fig. 5 SHAP dependence plots of key features in CHD risk prediction. **A** SHAP feature dependence plots (RC & Smoking). **B** SHAP feature dependence plots (FPG & Drinking). **C** SHAP feature dependence plots (MLR & PLR). **D** SHAP feature dependence plots (BMI & Drinking). **E** SHAP feature dependence plots (NLR & TyG). **F** SHAP feature dependence plots demonstrating the effect of PLR. **G** SHAP feature dependence plots (Drinking & FPG). **H** SHAP feature dependence plots of (TyG & smoking). **I** SHAP feature dependence plots demonstrating the effect of smoking. SHAP, shapley additive explanations; TyG, triglyceride-glucose index; RC, residual cholesterol; NLR, neutrophil-to-lymphocyte ratio; MLR, monocyte-to-lymphocyte ratio; PLR, platelet-to-lymphocyte ratio; BMI, body mass index; FPG, fasting plasma glucose. **Note:** For binary variables (Smoking, Drinking), 0 = no and 1 = yes

controlled levels of LDL-C, suggesting that RC may represent a “residual risk” after statin therapy [17, 23, 24]. The SHAP interaction analysis in this study shows that smoking status significantly moderates the predictive effect of RC, and that the pathogenic effect of RC is more pronounced in smokers. Smoking enhances lipoprotein modification through oxidative stress and inflammatory activation, accelerates oxidative modification of RC particles, and promotes foam cell formation, endothelial dysfunction, and vascular inflammation [25, 26]. This metabolic-behavioral synergy provides important clues to understanding the complex etiology of “brain-heart co-morbidity” and highlights the need for concurrent lipid management and smoking cessation interventions in patients with carotid stenosis.

In this study, the association between fasting plasma glucose (FPG) and CHD risk is significantly modulated by the degree of carotid stenosis. The RCS curves show that patients with severe stenosis exhibit higher risk sensitivity in the hypoglycemic range, whereas patients with mild-to-moderate stenosis show a steeper increase in risk in the hyperglycemic range (P for interaction = 0.036).

This interaction effect may reflect the differences in the vascular biological basis of different stenosis stages: in patients with mild-to-moderate stenosis, the vessel wall still has certain structural and functional reserves, and high glucose-induced inflammation, oxidative stress, and endothelial damage are more likely to directly trigger the destabilization of atherosclerotic plaques, thus accelerating CHD events. In contrast, at the stage of severe stenosis, plaques are usually highly inflamed and destabilized, and the risk is at an elevated plateau. Thus, they no longer show the same magnitude of sensitivity to further hyperglycemic stimuli [27]. Although the RCS analysis suggested a significant interaction, the SHAP subgroup distribution test did not reach statistical significance ($P = 0.203$), suggesting that the smoothing properties of the machine learning model when dealing with non-monotonic relationships may mask some of the true interaction effects [28]. In addition, the SHAP dependency plot shows that alcohol consumption has a moderating effect on the FPG effect: moderate drinkers have a reduced risk in the moderate glycemic range. This trend is consistent with the observation from a Chinese prospective cohort that

moderate alcohol consumption may have a protective effect against diabetes [29]. Moderate alcohol consumption may have beneficial effects on glucose metabolism by improving insulin sensitivity, but excessive alcohol consumption exacerbates metabolic disturbances by impairing liver function, inducing inflammatory responses, and interfering with glucose homeostasis [30]. This bidirectional effect may explain the complex pattern observed in the SHAP interaction analysis. It is worth noting, however, that alcohol consumption may simultaneously increase the risk of other health outcomes such as cancer, and therefore caution is still warranted with regard to alcohol consumption recommendations for the overall population. Based on these findings, differential glycemic management strategies may be considered: patients with mild-to-moderate stenosis should have their FPG more tightly controlled within the normal range to reduce the risk associated with hyperglycemia, whereas patients with severe stenosis may be able to accept relatively loose glycemic targets in the presence of a highly restricted hematological reserve, and measures such as revascularization should be prioritized to improve perfusion.

Triglyceride-glucose index (TyG), as an integrating indicator of FPG and triglycerides, occupies an important position in the model. It is considered a simple alternative marker of insulin resistance [31]. Insulin resistance not only promotes atherogenic dyslipidemia (high triglycerides, low HDL-C, and increased number of small and dense LDL particles), but also directly damages the vasculature by activating the sympathetic nervous system, promoting sodium retention, and causing endothelial dysfunction [32, 33]. Thus, there is a clear pathophysiological basis for its producing higher positive SHAP values for model output in the high TyG region. One study quantified the effect of insulin resistance on lesion severity by showing an increase of 2.44 points in the coronary Gensini score for every 1-unit increase in TyG index through multivariate logistic regression analysis [34]. The Chinese RCSCD-TCM study further revealed that the correlation between TyG index and carotid plaque was more pronounced in women, middle-aged patients, and diabetic subgroups [16], and this heterogeneity may be related to the modulation of insulin sensitivity by sex hormones, age-related metabolic changes, and amplification of the effect of hyperglycemia on insulin resistance. In this study, female patients constituted the majority of the study population (82.16%), resulting in a limited number of male patients in the validation set ($n = 27$) following random partitioning, which led to a statistically significant difference in sex distribution between the training and validation sets (79.88% vs. 87.50%, $P = 0.019$). Nevertheless, sex was not identified as an important predictive feature in the final model, and all remaining baseline variables were well-balanced

between the two sets (all $P > 0.05$), indicating a low risk of systematic bias. Future studies should expand sample sizes and incorporate more sex-balanced cohorts to further validate the generalizability of the model. We found that smokers are more inclined to exhibit high SHAP values at higher levels of TyG. Smoking may exacerbate insulin resistance and metabolic stress through pathways that promote oxidative stress, enhance inflammatory cascade responses, increase sympathetic nerve activity, and promote platelet adhesion [35]. Thus, in the context of elevated TyG, smoking may have a superimposed effect on metabolic abnormalities, making individuals more susceptible to a state of accelerated development of vascular endothelial damage and atherosclerosis, thereby increasing the predictive value of CHD risk. This trend provides model-level evidence for a possible interaction between metabolic disorders and adverse lifestyle factors.

Neutrophil-to-lymphocyte ratio (NLR) shows a typical saturation curve in the SHAP dependence plot, with a rapid rise in SHAP values in the low value interval followed by a plateau phase. The pathophysiological significance of NLR lies in its integration of the homeostatic state of pro-inflammatory innate immunity (neutrophils) and anti-inflammatory adaptive immunity (lymphocytes). Neutrophils are directly involved in the process of plaque destabilization through the release of myeloperoxidase (MPO), matrix metalloproteinases (MMPs), and neutrophil extracellular traps (NETs) [36]. Clinical studies have confirmed that elevated NLR is closely associated with coronary plaque rupture, incidence of acute coronary syndromes, and poor prognosis after myocardial infarction [13, 37, 38]. At the same time, lymphopenia reflects impaired adaptive immunoregulation, particularly a decline in the number or function of regulatory T cells (Tregs), leading to a tilt of the pro-inflammatory/anti-inflammatory balance in the pro-inflammatory direction [39]. SHAP interaction analysis shows that triglyceride levels modulate the predictive effect of NLR. This finding reveals the importance of the metabolic-inflammatory axis: hypertriglyceridemia amplifies the inflammatory response through multiple mechanisms, including promotion of activation of endothelial cells by lipolytic products of triglyceride-rich lipoproteins, enhancement of leukocyte adhesion, and induction of proinflammatory cytokine release [40]. This synergistic effect may explain the nonlinear increase in cardiovascular risk in patients with metabolic syndrome.

Monocyte-to-lymphocyte ratio (MLR) similarly shows a saturated non-linear relationship. Monocytes are the core effector cells of atherosclerosis, and their recruitment to the vessel wall and differentiation into macrophages and foam cells are key steps in plaque formation and progression [41]. A study including 12,093 patients confirmed that MLR was significantly associated with the

incidence, number, and echogenicity of carotid plaques, particularly in identifying hypoechoic plaques, an ultrasound feature of vulnerable plaques [42]. This finding has important clinical implications as hypoechoic plaques typically have larger necrotic cores, thinner fibrous caps, and higher risk of rupture [43]. A prospective study by Kalhor et al. shows that elevated MLR is independently associated with major adverse cardiovascular events, suggesting that MLR may not only reflect the current inflammatory state, but also predict future cardiovascular events [13]. SHAP interaction analyses show that PLR modulates the predictive effect of MLR, suggesting a complex interaction between leukocyte subpopulations.

Monocytes and platelets can form aggregates via the P-selectin-PSGL-1 pathway, which promotes monocyte activation, tissue factor expression, and polarization of pro-inflammatory phenotypes [44]. This intercellular crosstalk accelerates the thromboinflammatory cascade response and may underlie the cytological basis of the MLR-PLR interaction. PLR exhibits a distinctive U- or J-shaped curve, with the lowest SHAP values in the range of approximately 100–150; either too low or too high values are associated with an increased risk of CHD. PLR exhibits the most significant between-group differences in the distribution of SHAP values when compared by stratification of the degree of stenosis, which may reflect the key pathological role of the platelet-inflammation axis in the severe stenosis stage. Elevated PLR reflects the dual abnormality of platelet activation and relative reduction of lymphocytes. Platelets are not only the core cells of thrombosis, but recent studies have also revealed their important immunomodulatory functions: chemokines (e.g., PF4, RANTES) and growth factors (e.g., PDGF, TGF- β) released by activated platelets promote leukocyte recruitment, vascular smooth muscle cell proliferation, and extracellular matrix remodeling [45, 46]. A high PLR has been shown to be an independent marker of major in-hospital adverse cardiovascular events and the severity of coronary artery disease in patients with acute myocardial infarction [14]. However, a low PLR is also indicative of a poor prognosis, which may reflect a state of severely impaired immune function or malnutrition [47]. A relative increase in lymphocytes but a decrease in PLR may suggest chronic inflammatory depletion, hematopoietic dysfunction, or immune senescence. This complex bidirectional association underscores the unique value of PLR as an integrative biomarker and suggests that it needs to be interpreted in the context of specific clinical situations in clinical applications.

BMI shows an irregular fluctuation pattern in the SHAP dependence plot, which does not show the classical linear or monotonic nonlinear relationship. This phenomenon may be related to the “obesity paradox” that has been widely discussed in cardiovascular disease

research: several studies have found that overweight or mildly obese cardiovascular disease patients have a better prognosis than normal-weight or low-weight patients [48]. However, the explanation for this phenomenon is controversial and may involve selection bias, reverse causality (disease-induced weight loss), and metabolic health heterogeneity. The deeper reason lies in the intrinsic limitations of BMI: its inability to distinguish between muscle mass, adiposity, fat distribution, and metabolic health status. “Metabolically healthy obese” individuals have elevated BMI but maintain insulin sensitivity, normal lipid metabolism, and lower levels of inflammation, with no significant increase in cardiovascular risk. In contrast, “metabolically unhealthy normal weight” individuals have normal BMI but have elevated cardiovascular risk due to the presence of visceral fat accumulation, insulin resistance, and chronic inflammation [49]. This metabolic phenotypic heterogeneity may lead to a complex nonlinear relationship between BMI and CHD risk. Alcohol consumption as an interaction variable for BMI further adds to the complexity of the relationship. Alcohol consumption affects energy metabolism, appetite regulation, and fat distribution, and its moderating effect on the BMI-CHD relationship may vary depending on the amount of alcohol consumed, the pattern of alcohol consumption, and individual differences [50]. It is noteworthy that the TyG index incorporated into this model serves as an alternative marker for insulin resistance, aiding in distinguishing individuals with similar BMIs but markedly different metabolic statuses. This partially explains the complex nonlinear relationship between BMI and coronary heart disease risk. Future studies incorporating additional indicators of body fat distribution, such as waist circumference or waist-to-hip ratio, could enhance the precision of identifying high-risk metabolic phenotypes and improve risk stratification efficacy [51].

This study developed a random forest model integrating nine clinical features to predict the risk of CHD in patients with carotid artery stenosis. Unlike previous studies primarily relying on traditional cardiovascular risk factors, this model incorporates metabolic and inflammatory biomarkers based on the SHAP feature selection method, thereby providing a more comprehensive characterization of the underlying pathophysiological network of the disease. The superiority of this multidimensional integration strategy is demonstrated not only by improved discriminative performance but also by robust biological plausibility. Positioned as a clinical decision support tool rather than an independent diagnostic method, this model can be combined with existing risk assessment systems like the Framingham risk score to achieve more refined risk stratification management [52, 53]. The selection of probability cutoff

values should be flexibly adjusted based on specific clinical contexts: in scenarios prioritizing minimized missed diagnoses, lower cutoff values should be adopted to enhance sensitivity; conversely, in resource-constrained settings requiring control of unnecessary testing, cutoff values may be appropriately raised to ensure specificity. The optimal setting of these cutoff values in real-world clinical environments awaits further validation through prospective studies.

Limitations

However, this study has several noteworthy limitations. First, relying on ultrasound for stenosis grading and plaque classification introduces potential variability compared with histology- or MRI-based assessments. In addition, due to the high false-positive rate for diagnosis of coronary artery disease by coronary CTA compared with coronary DSA, the positive predictive value (PPV) of the model may be underestimated, potentially affecting the objective assessment of its clinical utility. Future research should prioritize the inclusion of cases confirmed by digital subtraction angiography (DSA) to further enhance the accuracy of the diagnostic reference standard and conduct more rigorous validation of the model's predictive performance. Second, The associations revealed by SHAP dependency plots cannot directly imply causality, and the observed interaction patterns may be susceptible to unmeasured confounding factors, including biochemical variables not captured in the present dataset such as glycated hemoglobin (HbA1c), high-sensitivity C-reactive protein (hs-CRP), and lipoprotein(a), as well as concomitant medication use. In particular, the use of statins and antiplatelet agents may directly modify key lipid parameters such as RC and LDL-C, and platelet or inflammatory markers such as PLR, potentially confounding the observed associations and limiting the interpretability of the interaction effects identified. Third, the machine-learning model needs to be validated in an external cohort to assess its generalizability and clinical utility. Fourth, this study used a cross-sectional design and was unable to assess the temporal relationship between dynamic changes in biomarkers and disease progression. Finally, We excluded samples with incomplete data, which may introduce some selection bias. And the statistical power of subgroup analyses was insufficient, limited by the sample size; some of the interaction tests may have a risk of type II error, thus failing to further explore the heterogeneity of effects in refined subgroups, such as plaque stability and extent of lesion involvement. In the future, it is necessary to expand the sample size and integrate imaging and multi-omics data to conduct prospective cohort studies to verify the long-term prognostic value.

Conclusions

In this study, we developed and validated an ML prediction model to identify the risk of CHD in patients with carotid artery stenosis by combining relevant risk factors. The RF model demonstrated excellent and reliable predictive performance. Combining metabolic and immunoinflammatory predictors to build a prediction model can significantly improve the prediction performance. Therefore, we believe that our study can be used for individualized prediction of the risk of CHD in patients with carotid artery stenosis, thereby assisting clinicians in performing early and precise interventions to effectively reduce or delay the development of stroke and myocardial infarction. However, external validation is needed to demonstrate the accuracy of model predictions in future studies.

Abbreviations

AISI	Aggregate index of systemic inflammation
AUC	Area under the curve
BMI	Body mass index
Boruta	Boruta feature selection algorithm
CHD	Coronary heart disease
DCA	Decision curve analysis
DSA	Digital subtraction angiography
dNLR	Derived neutrophil-to-lymphocyte ratio
FPG	Fasting plasma glucose
GBM	Gradient boosting machine
HDL-C	High-density lipoprotein cholesterol
KNN	K-nearest neighbors
LASSO	Least absolute shrinkage and selection operator
LDA	Linear discriminant analysis
LDL-C	Low-density lipoprotein cholesterol
LHR	Lymphocyte-to-HDL-C ratio
LR	Logistic regression
MDA	Mean decrease accuracy
MDG	Mean decrease Gini
MHR	Monocyte-to-HDL-C ratio
ML	Machine learning
MLR	Monocyte-to-lymphocyte ratio
NHR	Neutrophil-to-HDL-C ratio
NLR	Neutrophil-to-lymphocyte ratio
NN	Neural network
NPV	Negative predictive value
PDGF	Platelet-derived growth factor
PF4	Platelet factor 4
PHR	Platelet-to-HDL-C ratio
PLS	Partial least squares
PLR	Platelet-to-lymphocyte ratio
PLT	Platelet count
PNR	Platelet-to-neutrophil ratio
PPV	Positive predictive value
RBC	Red blood cell count
RC	Residual cholesterol
RCS	Restricted cubic spline
RF	Random forest
RF-MDA	Random forest-mean decrease accuracy
RF-MDG	Random forest-mean decrease Gini
ROC	Receiver operating characteristic
SHAP	SHapley additive explanations
SII	Systemic immune-inflammation index
SIRI	Systemic inflammation response index
SR-BS	Stepwise regression-backward selection
SVM	Support vector machine
SVM-RFE	Support vector machine-recursive feature elimination
TC	Total cholesterol

TGF- β	Transforming growth factor beta
TG	Triglycerides
THR	Triglyceride-to-HDL-C ratio
TyG	Triglyceride-glucose index
VIF	Variance inflation factor
WBC	White blood cell count
XGBoost	Extreme gradient boosting

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12872-026-05766-0>.

Supplementary Material 1.

Authors' contributions

LWZ, GKM and ZHH conceived the idea for the study and wrote the manuscript. XWD and BB helped analyze the study data and revise the manuscript. TXG and WHG directed the writing and analysis of this article.

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Data availability

The original contributions of this study are included in the article/ supplementary material. Further inquiries can be directed to the corresponding author.

Declarations

Ethics approval and consent to participate

The study was approved by the Tianjin Huanhu Hospital (Approval No. 2026-045), which waived the need for written informed consent due to the retrospective study design.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Martin SS, Aday AW, Allen NB, Almarzooq ZI, Anderson CAM, Arora P, Avery CL, Baker-Smith CM, Bansal N, Beaton AZ, et al. 2025 Heart Disease and Stroke Statistics: A Report of US and Global Data From the American Heart Association. *Circulation*. 2025;151(8):e41–660.
- Björkegren JLM, Lusis AJ. Atherosclerosis: Recent developments. *Cell*. 2022;185(10):1630–45.
- Pi H, Wang G, Wang Y, Zhang M, He Q, Zheng X, Yin K, Zhao G, Jiang T. Immunological perspectives on atherosclerotic plaque formation and progression. *Front Immunol*. 2024;15:1437821.
- Steven S, Frenis K, Oelze M, Kalinovic S, Kuntic M, Bayo Jimenez MT, Vujacic-Mirski K, Helmstädter J, Kröller-Schön S, Münzel T et al. Vascular Inflammation and Oxidative Stress: Major Triggers for Cardiovascular Disease. *Oxid Med Cell Longev* 2019, 2019:7092151.
- Roy P, Orecchioni M, Ley K. How the immune system shapes atherosclerosis: roles of innate and adaptive immunity. *Nat Rev Immunol*. 2022;22(4):251–65.
- Bos D, Arshi B, van den Bouwhuisen QJA, Ikram MK, Selwaness M, Vernooij MW, Kavousi M, van der Lugt A. Atherosclerotic Carotid Plaque Composition and Incident Stroke and Coronary Events. *J Am Coll Cardiol*. 2021;77(11):1426–35.
- Yao EY, Ge Z, Huo M, Huang G, Tang J, Liu Y, Tan Z, Zeng Z, Cao Y. Development and validation of a machine learning model for predicting vulnerable carotid plaques using routine blood biomarkers and derived indicators: insights into sex-related risk patterns. *Cardiovasc Diabetol*. 2025;24(1):326.
- Song YB, Arbab-Zadeh A, Matheson MB, Ostovaneh MR, Vavere AL, Dewey M, Rochitte C, Niinuma H, Laham R, Schuijff JD, et al. Contemporary Discrepancies of Stenosis Assessment by Computed Tomography and Invasive Coronary Angiography. *Circ Cardiovasc Imaging*. 2019;12(2):e007720.
- Soehnlein O, Libby P. Targeting inflammation in atherosclerosis - from experimental insights to the clinic. *Nat Rev Drug Discov*. 2021;20(8):589–610.
- Cui C, Liu L, Qi Y, Han N, Xu H, Wang Z, Shang X, Han T, Zha Y, Wei X, et al. Joint association of TyG index and high sensitivity C-reactive protein with cardiovascular disease: a national cohort study. *Cardiovasc Diabetol*. 2024;23(1):156.
- Shen L, Zhai W, Jiang P, Liang F, Li R, Xu D, Zhang Q, Zhang J, Tao X. Systemic inflammation modulates lipoprotein(a)-associated coronary stenosis in the chronic coronary syndromes. *Am J Prev Cardiol*. 2025;24:101324.
- Mahdavi-Roshan M, Mozafarihashjin M, Shoaibinobarian N, Ghorbani Z, Salari A, Savarraksh A, Hekmatdoost A. Evaluating the use of novel atherogenicity indices and insulin resistance surrogate markers in predicting the risk of coronary artery disease: a case-control investigation with comparison to traditional biomarkers. *Lipids Health Dis*. 2022;21(1):126.
- Kalhor P, Mahmoudi Hamidabad N, Hellou E, Nogami K, Manzato M, Lerman LO, Lerman A. Association of monocyte-to-lymphocyte ratio with coronary endothelial dysfunction and cardiovascular events in patients with angina and nonobstructive coronary artery disease. *Atherosclerosis*. 2025;410:120539.
- Li XT, Fang H, Li D, Xu FQ, Yang B, Zhang R, An Y. Association of platelet to lymphocyte ratio with in-hospital major adverse cardiovascular events and the severity of coronary artery disease assessed by the Gensini score in patients with acute myocardial infarction. *Chin Med J (Engl)*. 2020;133(4):415–23.
- Bagyura Z, Kiss L, Lux Á, Csobay-Novák C, Jermendy ÁL, Polgár L, Tabák ÁG, Soós P, Szelid Z, Merkely B, et al. Neutrophil-to-Lymphocyte Ratio Is an Independent Risk Factor for Coronary Artery Disease in Central Obesity. *Int J Mol Sci*. 2023;24(8):7397.
- Su J, Li Z, Huang M, Wang Y, Yang T, Ma M, Ni T, Pan G, Lai Z, Li C, et al. Triglyceride glucose index for the detection of the severity of coronary artery disease in different glucose metabolic states in patients with coronary heart disease: a RCSCD-TCM study in China. *Cardiovasc Diabetol*. 2022;21(1):96.
- Hao QY, Gao JW, Yuan ZM, Gao M, Wang JF, Schiele F, Zhang SL, Liu PM. Remnant Cholesterol and the Risk of Coronary Artery Calcium Progression: Insights From the CARDIA and MESA Study. *Circ Cardiovasc Imaging*. 2022;15(7):e014116.
- Moon JH, Kim Y, Oh TJ, Moon JH, Kwak SH, Park KS, Jang HC, Choi SH, Cho NH. Triglyceride-Glucose Index Predicts Future Atherosclerotic Cardiovascular Diseases: A 16-Year Follow-up in a Prospective, Community-Dwelling Cohort Study. *Endocrinol Metab (Seoul)*. 2023;38(4):406–17.
- Handelman GS, Kok HK, Chandra RV, Razavi AH, Lee MJ, Asadi H. eDoc-tor: machine learning and the future of medicine. *J Intern Med*. 2018;284(6):603–19.
- Deo RC. Machine Learning in Medicine. *Circulation*. 2015;132(20):1920–30.
- Johnson KW, Torres Soto J, Glicksberg BS, Shameer K, Miotto R, Ali M, Ashley E, Dudley JT. Artificial Intelligence in Cardiology. *J Am Coll Cardiol*. 2018;71(23):2668–79.
- Brinjikji W, Huston J 3rd, Rabinstein AA, Kim GM, Lerman A, Lanzino G. Contemporary carotid imaging: from degree of stenosis to plaque vulnerability. *J Neurosurg*. 2016;124(1):27–42.
- Castañer O, Pintó X, Subirana I, Amor AJ, Ros E, Hernández Á, Martínez-González M, Corella D, Salas-Salvadó J, Estruch R, et al. Remnant Cholesterol, Not LDL Cholesterol, Is Associated With Incident Cardiovascular Disease. *J Am Coll Cardiol*. 2020;76(23):2712–24.
- Drexel H, Mader A, Larcher B, Festa A, Vonbank A, Fraunberger P, Leiberer A, Saely CH. Remnant cholesterol and long-term incidence of death in coronary artery disease patients. *Atherosclerosis*. 2025;401:119048.

25. Münzel T, Hahad O, Kuntic M, Keaney JF, Deanfield JE, Daiber A. Effects of tobacco cigarettes, e-cigarettes, and waterpipe smoking on endothelial function and clinical outcomes. *Eur Heart J*. 2020;41(41):4057–70.
26. Buchanan ND, Grimmer JA, Tanwar V, Schwieterman N, Mohler PJ, Wold LE. Cardiovascular risk of electronic cigarettes: a review of preclinical and clinical studies. *Cardiovasc Res*. 2020;116(1):40–50.
27. Juricic S, Klac J, Stojkovic S, Tesic M, Jovanovic I, Aleksandric S, Dobric M, Zivkovic S, Maricic B, Simeunovic D et al. Molecular and Pathophysiological Mechanisms Leading to Ischemic Heart Disease in Patients with Diabetes Mellitus. *Int J Mol Sci*. 2025;26(9):3924.
28. Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, Katz R, Himelfarb J, Bansal N, Lee SI. From Local Explanations to Global Understanding with Explainable AI for Trees. *Nat Mach Intell*. 2020;2(1):56–67.
29. Lv J, Yu C, Guo Y, Bian Z, Yang L, Chen Y, Hu X, Hou W, Chen J, Chen Z, et al. Adherence to a healthy lifestyle and the risk of type 2 diabetes in Chinese adults. *Int J Epidemiol*. 2017;46(5):1410–20.
30. Roerecke M. Alcohol's Impact on the Cardiovascular System. *Nutrients*. 2021;13(10):3419.
31. Khalaji A, Behnouth AH, Khanmohammadi S, Ghanbari Mardasi K, Sharifkashani S, Sahebkar A, Vinciguerra C, Cannavo A. Triglyceride-glucose index and heart failure: a systematic review and meta-analysis. *Cardiovasc Diabetol*. 2023;22(1):244.
32. Apostolopoulou M, Lambadiari V, Roden M, Dimitriadis GD. Insulin Resistance in Type 1 Diabetes: Pathophysiological, Clinical, and Therapeutic Relevance. *Endocr Rev*. 2025;46(3):317–48.
33. Accili D, Deng Z, Liu Q. Insulin resistance in type 2 diabetes mellitus. *Nat Rev Endocrinol*. 2025;21(7):413–26.
34. Zhao J, Fan H, Wang T, Yu B, Mao S, Wang X, Zhang W, Wang L, Zhang Y, Ren Z, et al. TyG index is positively associated with risk of CHD and coronary atherosclerosis severity among NAFLD patients. *Cardiovasc Diabetol*. 2022;21(1):123.
35. Chen Z, Liu XA, Kenny PJ. Central and peripheral actions of nicotine that influence blood glucose homeostasis and the development of diabetes. *Pharmacol Res*. 2023;194:106860.
36. Döring Y, Soehnlein O, Weber C. Neutrophil Extracellular Traps in Atherosclerosis and Atherothrombosis. *Circ Res*. 2017;120(4):736–43.
37. Dziedzic EA, Gaşior JS, Tuzimek A, Dąbrowski M, Jankowski P. Neutrophil-to-Lymphocyte Ratio Is Not Associated with Severity of Coronary Artery Disease and Is Not Correlated with Vitamin D Level in Patients with a History of an Acute Coronary Syndrome. *Biology (Basel)*. 2022;11(7):1001.
38. Jiang J, Zeng H, Zhuo Y, Wang C, Gu J, Zhang J, Zhang H. Association of Neutrophil to Lymphocyte Ratio With Plaque Rupture in Acute Coronary Syndrome Patients With Only Intermediate Coronary Artery Lesions Assessed by Optical Coherence Tomography. *Front Cardiovasc Med*. 2022;9:770760.
39. Ketelhuth DF, Hansson GK. Adaptive Response of T and B Cells in Atherosclerosis. *Circ Res*. 2016;118(4):668–78.
40. Xu D, Xie L, Cheng C, Xue F, Sun C. Triglyceride-rich lipoproteins and cardiovascular diseases. *Front Endocrinol (Lausanne)*. 2024;15:1409653.
41. Tabas I, Bornfeldt KE. Macrophage Phenotype and Function in Different Stages of Atherosclerosis. *Circ Res*. 2016;118(4):653–67.
42. Ma M, Liu Y, Wang L, Yang R, Li Z, Gao S, Li L, Yu C. Relationship Between Monocyte-to-Lymphocyte Ratio as Well as Other Leukocyte-Derived Ratios and Carotid Plaques in Patients with Coronary Heart Disease: A RCSCD-TCM Study. *J Inflamm Res*. 2022;15:5141–56.
43. Paraskevas KI, Brown MM, Lal BK, Myrcha P, Lyden SP, Schneider PA, Poredos P, Mikhailidis DP, Secemsky EA, Musialek P, et al. Recent advances and controversial issues in the optimal management of asymptomatic carotid stenosis. *J Vasc Surg*. 2024;79(3):695–703.
44. Wong DJ, Park DD, Park SS, Haller CA, Chen J, Dai E, Liu L, Mandhapat AR, Eradi P, Dhakal B, et al. A PSGL-1 glycomimetic reduces thrombus burden without affecting hemostasis. *Blood*. 2021;138(13):1182–93.
45. Schroer AB, Ventura PB, Sucharov J, Misra R, Chui MKK, Bieri G, Horowitz AM, Smith LK, Encabo K, Tenggara I, et al. Platelet factors attenuate inflammation and rescue cognition in ageing. *Nature*. 2023;620(7976):1071–9.
46. Girdhar K, Mine K, DaCosta JM, Atkinson MA, Ludvigsson J, Altindis E. Sex-specific cytokine, chemokine, and growth factor signatures in T1D patients and progressors. *Faseb j*. 2024;38(24):e70270.
47. Ruan GT, Ge YZ, Xie HL, Hu CL, Zhang Q, Zhang X, Tang M, Song MM, Zhang XW, Liu T, et al. Association Between Systemic Inflammation and Malnutrition With Survival in Patients With Cancer Sarcopenia-A Prospective Multicenter Study. *Front Nutr*. 2021;8:811288.
48. Tutor AW, Lavie CJ, Kachur S, Milani RV, Ventura HO. Updates on obesity and the obesity paradox in cardiovascular diseases. *Prog Cardiovasc Dis*. 2023;78:2–10.
49. Schulze MB, Stefan N. Metabolically healthy obesity: from epidemiology and mechanisms to clinical implications. *Nat Rev Endocrinol*. 2024;20(11):633–46.
50. Peng H, Wang S, Wang M, Wang X, Guo H, Huang J, Wu T. Lifestyle Factors, Genetic Risk, and Cardiovascular Disease Risk among Breast Cancer Survivors: A Prospective Cohort Study in UK Biobank. *Nutrients*. 2023;15(4):864.
51. Li Y, He Y, Yang L, Liu Q, Li C, Wang Y, Yang P, Wang J, Chen Z, Huang X. Body Roundness Index and Waist-Hip Ratio Result in Better Cardiovascular Disease Risk Stratification: Results From a Large Chinese Cross-Sectional Study. *Front Nutr*. 2022;9:801582.
52. Sud M, Sivaswamy A, Austin PC, Abdel-Qadir H, Anderson TJ, Naimark DMJ, Lee DS, Roifman I, Thanassoulis G, Tu K, et al. Implications of five different risk models in primary prevention guidelines. *Eur Heart J Qual Care Clin Outcomes*. 2025;11(4):388–96.
53. Kasim SS, Ibrahim N, Malek S, Ibrahim KS, Aziz MF, Song C, Chia YC, Ramli AS, Negishi K, Mat Nasir N. Validation of the general Framingham Risk Score (FRS), SCORE2, revised PCE and WHO CVD risk scores in an Asian population. *Lancet Reg Health West Pac*. 2023;35:100742.

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